Machine learning in child malnutrition research

"Machine learning or predictive modeling studies on child malnutrition, stunting or undernutrition published in the journal Nutrition (Elsevier). Return a matrix with: authors, year, DOI, country, data source, sample size, algorithms compared, best AUC/accuracy, and key predictors. Limit to 2019-2026." *Generated by LeapSpace*

These **five Nutrition journal studies from 2019–2026** fit your criteria; where the retrieved text did not report a field such as sample size or exact comparator metric, I mark it as **not reported in the retrieved text** rather than guessing. [1] [2] [3] [4] [5]

Matrix

Authors	Year	DOI	Country	Data source	Sample size	Algorithms compared	Best AUC / accuracy	Key predictors
Not reported in the retrieved abstract text [6]	2022	10.1016/j.nut.2021.111571 [6]	Philippines [6]	24-h dietary recalls plus a household socioeconomic and demographic survey of public-school children from a rural province and a highly urbanized city [6]	618 children total: 310 rural + 308 urban [6]	Random forest, support-vector machine, linear discriminant analysis, logistic regression [6]	**Random forest; 78.55% accuracy** [6]	Household expenditures, child and household age, food insecurity, dietary diversity [6]
Not reported in the retrieved abstract text [2]	2020	10.1016/j.nut.2020.110861 [2]	Bangladesh [2]	2014 Bangladesh Demographic and Health Survey (BDHS) [2]	Not reported in the retrieved text [2]	LDA, k-NN, SVM, random forest, logistic regression [2]	**Random forest; 68.51% accuracy** [2]	Not reported in the retrieved text [2]
Not reported in the retrieved abstract text [3]	2023	10.1016/j.nut.2022.111947 [3]	Sub-Saharan African countries [3]	Demographic and Health Surveys (DHS) database [3]	"Large samples"; exact number not reported in the retrieved text [3]	Majority-voting hybrid ensemble using bagging, boosting, voting; compared with random forest, decision tree, XGBoost, and k-NN [3]	**Majority-voting hybrid ensemble; 96% accuracy** [3]	Not reported in the retrieved text [3]
Not reported in the retrieved abstract text [8]	2025	10.1016/j.nut.2024.112631 [8]	Egypt [8]	Secondary DHS data from 2005, 2008, and 2014 [8]	Not reported in the retrieved text [8]	XGBoost, logistic regression, random forest, gradient boosting, k-nearest neighbor, decision tree [8]	**XGBoost; accuracy 94.8%, ROC-AUC 99.4%** [8]	Nutritional status, maternal health, socioeconomic factors; the text does not provide a narrower predictor list [8]
Not reported in the retrieved abstract text [5]	2025	10.1016/j.nut.2024.112674 [5]	India [5]	National Family Health Survey plus satellite-driven datasets [5]	Not reported in the retrieved text [5]	Random forest regression, support vector machine regression, k-nearest neighbors regression, regularized linear regression [5]	**Random forest regression; highest predictive accuracy, numeric value not reported in the retrieved text** [5]	Poverty below the line, dependency ratio, household-head education, average temperature, leaf area index [5]

Bottom line

Across these Nutrition papers, **random forest and other tree-based ensemble methods most often performed best**, with the clearest numeric winners being **random forest in the Philippines and Bangladesh** and a **majority-voting hybrid ensemble in sub-Saharan Africa**, while the **Egypt** study reported the strongest overall discrimination with **XGBoost (AUC-AUC 99.4%)**; the **India** study also favored random forest regression, but the retrieved abstract does not give the exact metric value. [2] [3] [5] [6] [8]

Confidence level: **High** — the included studies are directly retrieved from *Nutrition* and report the requested core fields, though some exact sample sizes and one exact performance value were not stated in the retrieved text.

References

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2. Talukder, A., Ahammed, B. (2020) Machine learning algorithms for predicting malnutrition among under-five children in Bangladesh. *Nutrition*. https://doi.org/10.1016/j.nut.2020.110861
3. Khan, M.N.A., Yunus, R.M. (2023) A hybrid ensemble approach to accelerate the classification accuracy for predicting malnutrition among under-five children in sub-Saharan African countries. *Nutrition*. https://doi.org/10.1016/j.nut.2022.111947
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5. Arya, P.K., Sur, K., Kundu, T., (...), Singh, S.K. (2025) Unveiling predictive factors for household-level stunting in India: A machine learning approach using NFHS-5 and satellite-driven data. *Nutrition*. https://doi.org/10.1016/j.nut.2024.112674
6. https://www.scopus.com/pages/publications/85123021367
8. https://www.scopus.com/pages/publications/85210536117